

Courtside: A Deployed Multi-Modal Sensing Instrument for Contextualizing Beach Volleyball Performance

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Abstract

Beach volleyball athletes experience large day-to-day performance variability that no single tool explains: wearables report overnight recovery without same-day context, weather services miss court-level microclimate (wind in particular), and mid-match self-report is impractical. We designed, built, and *deployed Courtside*, a multi-modal sensing instrument that fuses six streams—a custom court-side environmental node (ESP32-S3 with a DHT11 air temperature/humidity sensor, an MLX90614 infrared sand-surface thermometer, and an electret-microphone wind/audio proxy), wearable recovery physiology (Apple Watch via automated HealthKit export and Oura Ring via API), wearable workout intensity (in-session and prior-load), coarse regional weather (Open-Meteo), per-session subjective/objective performance logging, and the *spatial* and *relational* context of each session (which court, and which partners and opponents). The node streams over the athlete’s phone hotspot to a cloud backend (FastAPI + PostgreSQL on Render) and a mobile web app that runs live setup, session recording, evaluation entry, an interactive court map, and partner/opponent “personas” with statistically-guarded relationship signals. A four-stage factor-ranking analysis (correlation → sparse linear selection → tree ensembles → SHAP) closes the loop. We validate the instrument end-to-end on a single-athlete pilot—including a 20-respondent player survey that shaped the feature set and a live beach day capturing sand-surface temperatures near 40 °C against ~26 °C air, alongside synchronized Apple Watch recovery and in-session heart rate. We frame the contribution honestly: the value is the *working instrument*, the user-needs study, and the gap it fills—not statistical claims, which the small real sample ($N=2$ fully-instrumented sessions) cannot support. Where analytics need scale, we demonstrate them on clearly-labeled sample data *built around those two real sessions* to show the output the instrument produces.

1 Motivation

I began this project from my own experience. After starting to play beach volleyball at an elite collegiate level, I found it genuinely hard to tell whether I was improving, to discriminate the quality of one training session from another, and to know *what* actually drove a good or bad day. Playing with partners of all ages and skill levels, and across very different courts, made clear this is a common problem, not just mine: athletes know some days they play well and others they do not, but isolating *why* is hard because recovery, environment, game context, and mindset all interact, while existing tools each measure one dimension in isolation—or none at all.

Three blind spots. (1) *Wearables* (WHOOP, Oura, Garmin, Apple Watch) report readiness from overnight HRV and sleep but ignore same-day environmental load and subjective state. (2) *Weather*



Figure 1: The whole system in the field. The courtside node runs untethered on a tripod (battery-powered, streaming over the phone’s hotspot); the athlete reads live conditions in the web app while an Apple Watch supplies recovery and in-session heart rate. Everything in this photo is a working part of the deployed instrument.

apps report city-grid conditions and miss court-level microclimate: wind is reshaped by surrounding terrain—the Muir courts on the UCSD campus are ringed by buildings that create a swirling environment which reshapes serve-receive in a way no nearby weather station could capture—and sand-surface temperature varies with grain size, moisture, and direct sun in ways ambient air temperature cannot. (3) *Self-evaluation* is limited to what an athlete can remember and to their own pattern recognition; it carries heavy confounds (partner, opponent) and places the entire measurement burden on the athlete mid-competition, when compliance collapses.

The gap. No system fuses *recovery + environment + subjective state + performance*—together with *where* and *with whom* a session happened—for an individual beach volleyball athlete. Each stream exists in isolation.

Who benefits. The immediate end user is the recreational or collegiate beach athlete who today has no way to contextualize why a session felt the way it did. The broader health relevance is exertional-heat safety: beach volleyball is outdoors with no climate control, and early warning for heat strain requires exactly the physiological–environmental–subjective fusion this instrument produces. A sand-surface temperature near 40 °C—measured in our pilot—is well above the ~26 °C air temperature a weather app would report for the same moment.

2 Related Work

Commercial wearable recovery. WHOOP, Oura, and Garmin produce daily readiness scores from overnight biometrics; validation studies show reasonable resting HRV and sleep accuracy, but these scores are not contextualized against same-day environment or subjective state. Apple Watch exposes HRV (SDNN) and in-workout heart rate via HealthKit. Courtside treats these as one input among several rather than the whole picture.

Environmental monitoring in sport. The FIVB enforces WBGT-based heat protocols at professional events and the NATA publishes heat-illness position statements [1], but these rely on coarse point measurements and are largely absent from recreational and collegiate settings. We bring a low-cost court-level environmental node to exactly that gap.

Ecological momentary assessment (EMA). EMA is well-validated in sport and behavioral health [2]; voice-based capture with modern ASR has been proposed to lower burden. We scope voice EMA out (Section 8) in favor of a low-burden structured post-session form.

Wearables in beach volleyball. Prior work targets different questions: Marzano-Felisatti et al. (2024) [6] applied an electronic performance-tracking system to quantify physical-conditional performance of young male players, and the University of Sussex “Wearables in Beach Volleyball” project [7] investigates IMU-based match load and event detection. Both focus on movement classification and training-load quantification from a single modality. Neither fuses physiological recovery, environmental context, spatial/relational context, and subjective experience into a joint per-session analysis.

Gap. No published system performs this multi-modal, individual-athlete fusion for beach volleyball.

3 Project Aim

Primary aim. Build a working instrument that captures synchronized multi-modal sessions for a single beach volleyball athlete and runs a per-session factor-ranking analysis identifying which contextual streams plausibly explain performance variation. **End user:** the athlete; output is personally actionable, not a clinical or coaching product. **Non-goals:** not a deployable predictive model, not cross-sport, not clinically validated.

Following up on the proposal and update. Table 1 maps what was proposed and updated against what was delivered, with honest substitutions. The proposal committed to Oura + Apple Watch + a BME280 environmental node + a subjective rating + 6–10 sessions, with voice EMA and computer-vision (CV) kill/error extraction as stretch goals. The update reported the survey, bench-validated

Table 1: Proposed/updated vs. delivered. ✓ delivered, ~ substituted, + added beyond proposal, × de-scoped.

Component	Proposed/Update	Final
Wearable recovery (Oura)	API pull	✓ API
Wearable (Apple Watch)	HealthKit	✓ auto HealthKit export
In-session HR / workouts	stretch	✓ parsed (avg/max/energy)
Air temp / humidity	BME280	~ DHT11
Sand-surface temp (IR)	MLX90614	✓ MLX90614
Wind proxy (acoustic)	mic	✓ mic (qualitative)
Regional weather context	—	+ Open-Meteo
Web app / live capture	local logging	✓ cloud app + recording
Court map (spatial)	—	+ interactive map
Partner/opponent personas	—	+ relational analytics
Subjective outcome	1–10 rating	✓ rating (+ peer/coach)
4-stage analysis	planned	✓ implemented
Voice EMA	stretch	× de-scoped
CV kill/error outcome	stretch	× de-scoped
≥6 full-stream sessions	target	× small <i>N</i> (limitation)

sensing, and an MLX90614 in transit. For the final system, the environmental sensing, wearable ingestion, web app, and analysis pipeline were all delivered and deployed; the BME280 was substituted with a DHT11; the app additionally grew spatial (court map) and relational (partner/opponent personas) dimensions that were not in the proposal; and voice EMA and CV were de-scoped to keep the core instrument robust. The dominant gap versus the proposal is session count: collegiate season timing and hardware lead times compressed real data collection.

4 System Architecture and Implementation

The engineering is the heart of this project, so we describe the deployed system in detail. Figure 2 shows the end-to-end data path; it is a live system, not a mockup.

4.1 Courtside Environmental Node

An ESP32-S3 microcontroller (Fig. 3) reads three channels at 1 Hz: a **DHT11** (air temperature and humidity, one-wire digital); an **MLX90614** non-contact infrared thermometer (sand-surface temperature, I²C address 0x5A) that senses emitted infrared and therefore requires no contact and no exposed window (plastic blocks infrared, so the enclosure leaves an open aperture aimed at the sand); and an analog **electret microphone** whose 50 ms peak-to-peak amplitude serves as a low-cost wind/audio proxy. A deliberate hardware detail: the microphone is read on an **ADC1** pin, because the ESP32’s ADC2 is commandeered by the WiFi radio—reading the mic on ADC2 would silently fail whenever the radio is active.

The electronics are housed in a custom 3D-printed enclosure designed in Fusion 360 (Fig. 4) with a louvered vent for the air sensor, an open IR aperture, a foam-screened microphone port, and a 1/4-20 thread so the node mounts on a standard camera tripod and can be raised, leveled, and rotated on site. The node runs untethered from a USB power bank. Figure 3 shows the internals: the printed lid (left) carries custom standoffs and cutouts that seat each sensor at its aperture, while the body (right) holds the board, the tripod boss, and the sensor harness.

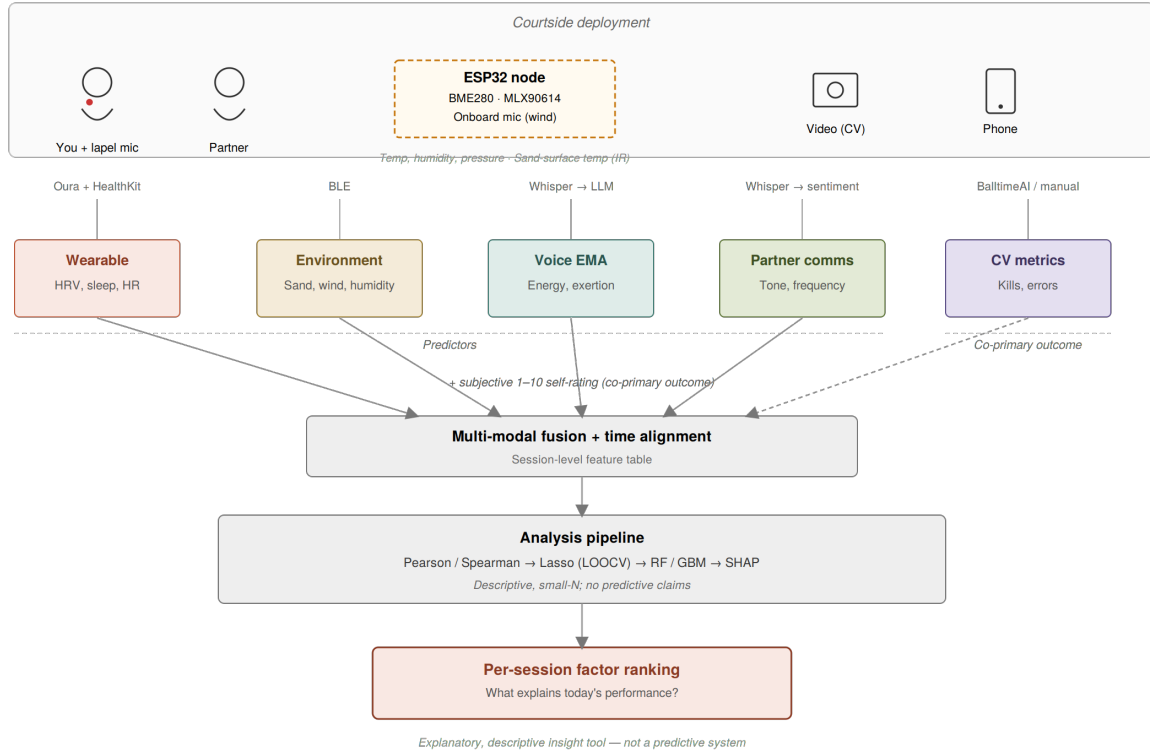


Figure 2: System architecture. The courtside node streams sensor readings over the athlete’s phone hotspot to a cloud backend; the Apple Watch and Oura push recovery and workout data to the same backend; the athlete logs subjective/objective outcomes from the mobile web app. All streams are time-aligned into one per-session feature table feeding the four-stage factor-ranking analysis.

4.2 Firmware and Connectivity

The firmware (C++/Arduino, built in PlatformIO) connects to the athlete’s iPhone Personal Hotspot as a WiFi station and HTTP-POSTs each reading as JSON to the backend, authenticated by a node key. It includes a boot-time I²C bus scan that prints discovered addresses for field debugging, and a serial-CSV fallback so the node still logs if connectivity drops. Because the ESP32-S3 is 2.4 GHz only, the hotspot is set to maximize-compatibility mode. This wireless-over-hotspot path is what lets the system run on a beach with no laptop and no SD card.

4.3 Cloud Backend and Data Model

A FastAPI service backed by PostgreSQL is deployed on Render and auto-deploys on each push. It exposes ingest endpoints (court readings, health, workouts), a session CRUD and recording API, people/personas, an aggregated court endpoint for the map, a relationship-signals endpoint, and a full export for the offline pipeline. The schema separates concerns into `court_readings`, `health_daily` (one row per day×source), `sessions`, `workouts`, and—added for the relational features—`people` and a `session_people` link table. Every wearable payload is retained losslessly in a raw JSON column so new signals can be promoted later without re-collection, and a

lightweight auto-migration adds new model columns to the live database on deploy.

4.4 Mobile Web App

The app (vanilla JS, no build step, installable as a phone web app) is the instrument’s user surface. Figure 5 shows it running live over cellular on the first instrumented beach day. It provides a *live setup* view with real-time per-sensor plots and a heuristic check (it warns when the IR reading $\approx$ air temperature, indicating the sensor is not yet aimed at the sand); a *recording* flow that bookmarks each session’s start/stop so its sensor time-series window is captured; and an *evaluation* form capturing the subjective 1–10 self-rating, optional peer/coach ratings, felt-wind, kills/errors, player state (energy, soreness, mental, warm-up, food timing), partner/opponents, location, notes, and geolocation.

4.5 Wearable and Weather Ingestion

Apple Watch data syncs automatically via the Health Auto Export app’s REST automation, posting HRV (SDNN), resting heart rate, sleep, respiratory rate, and *workouts* to the backend, upserted by date. Oura is pulled via its v2 API as a *second* recovery source, enabling cross-validation (Apple SDNN vs. Oura RMSSD). Open-Meteo [5] supplies coarse regional weather (wind, gust, temperature,

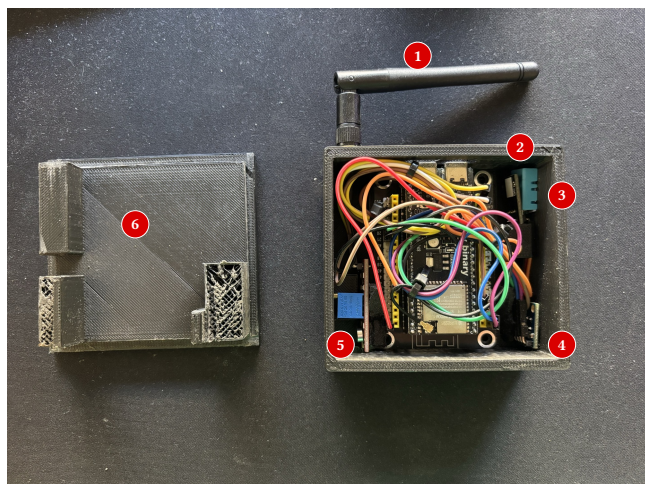


Figure 3: Inside the node. 1 ESP32-S3 antenna; 2 USB-C power; 3 DHT11 air temperature/humidity (top-right); 4 MLX90614 infrared sand-surface sensor (bottom-right); 5 electret microphone wind/audio proxy (bottom-left); 6 3D-printed lid with custom cutouts and standoffs that seat each sensor. (Callout positions per the build layout.)

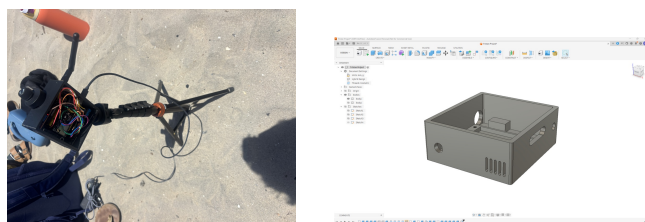


Figure 4: Left: the node deployed courtside on sand, tripod-mounted and battery-powered. Right: the enclosure CAD (Fusion 360)—louvered air-sensor vent, open IR aperture, and tripod mount.

humidity), kept clearly separate from on-court sensor readings—context, not mixed-in measurement.

4.6 Spatial Context: the Court Map

Because the app already captures geolocation, sessions are grouped by court and plotted on an interactive map (Fig. 6). Each marker is colored by the athlete’s average rating there; tapping a court reveals its session count, average rating, average node-measured wind and surface temperature, and who the athlete has played with at that court. This makes “I play better at La Jolla Shores than at Ocean Beach” a queryable, visual fact rather than a vague impression.

4.7 Relational Context: Partner/Opponent Personas

Beach volleyball is a 2v2 game and, candidly, a deeply relational and emotional one—who you play with and against shapes both performance and experience. Courtside therefore promotes partners and opponents from free text to first-class *people* records that accrue

history. For each person the app shows the athlete’s average rating *with* them (as partner) or *against* them (as opponent), a self-rated skill, and an *estimated* skill derived from results—a partner who consistently lifts the athlete’s rating, or an opponent who consistently lowers it, reads as stronger (Fig. 7, right).

Crucially, the app surfaces *relationship signals* as notifications (Fig. 7, left). For each person with enough sessions, it compares the athlete’s ratings with versus without that person using a Welch’s *t*-test (implemented in pure Python so the web service stays dependency-light), and flags notable, small-*N*-guarded effects: “you play your best with Maya,” or “tough matchup against Daniel.” This operationalizes the intuition every athlete has—“of course I played badly, I was up against the best player there”—into an explicit, evidence-weighted prompt. As with the pipeline, these analytics are shown on labeled sample data because the real pairing history is not yet large enough for inference.

5 Design Decisions and Development History

The system arrived at its present form through a sequence of deliberate decisions; we recount them as a narrative (ordered, undated) because the *reasoning* is the contribution.

Survey before silicon. Before investing further in hardware we ran a 20-player needs survey (Section 7) to decide what to sense. Wind ranking first is the reason the acoustic wind proxy survived and court noise and pace-of-play were dropped.

Sensor selection and an honest substitution. The proposal specified a BME280 (temperature, humidity, pressure). When the on-hand part proved unreliable, we substituted a DHT11 to keep air temperature and humidity working, accepting the loss of barometric pressure and lower resolution—a pragmatic trade we flag as a limitation rather than hide. The MLX90614 was chosen specifically because infrared sensing needs no ground contact and no fragile window.

Bench-first bring-up. Each sensor was validated on the bench before field use (Fig. 10, Fig. 8), and the firmware gained an I²C scanner so a missing sensor is diagnosable in seconds on the sand.

The pivotal pivot: from scripts to a cloud instrument. The original plan logged to a laptop over USB. We realized this is unusable on a beach, and re-architected around the athlete’s phone: the node streams over the phone hotspot to a cloud backend, and the phone itself is the live dashboard. This single decision—no laptop, no SD card—is what makes the instrument genuinely field-usable.

The app as the instrument, not an afterthought. Once the data lived in the cloud, the web app became the product surface: live setup, recording windows, and evaluation entry replaced a clipboard and a spreadsheet.

Frictionless wearable capture. Rather than manual exports, we adopted Health Auto Export’s REST automation so Apple Watch recovery and workouts arrive on their own, and kept Oura as an independent cross-check.

Spatial and relational dimensions last. With the core solid, we added the court map and the partner/opponent persona system—the features that most directly serve the “relationship sport” nature of beach volleyball—each backed by real database tables so they grow with use.

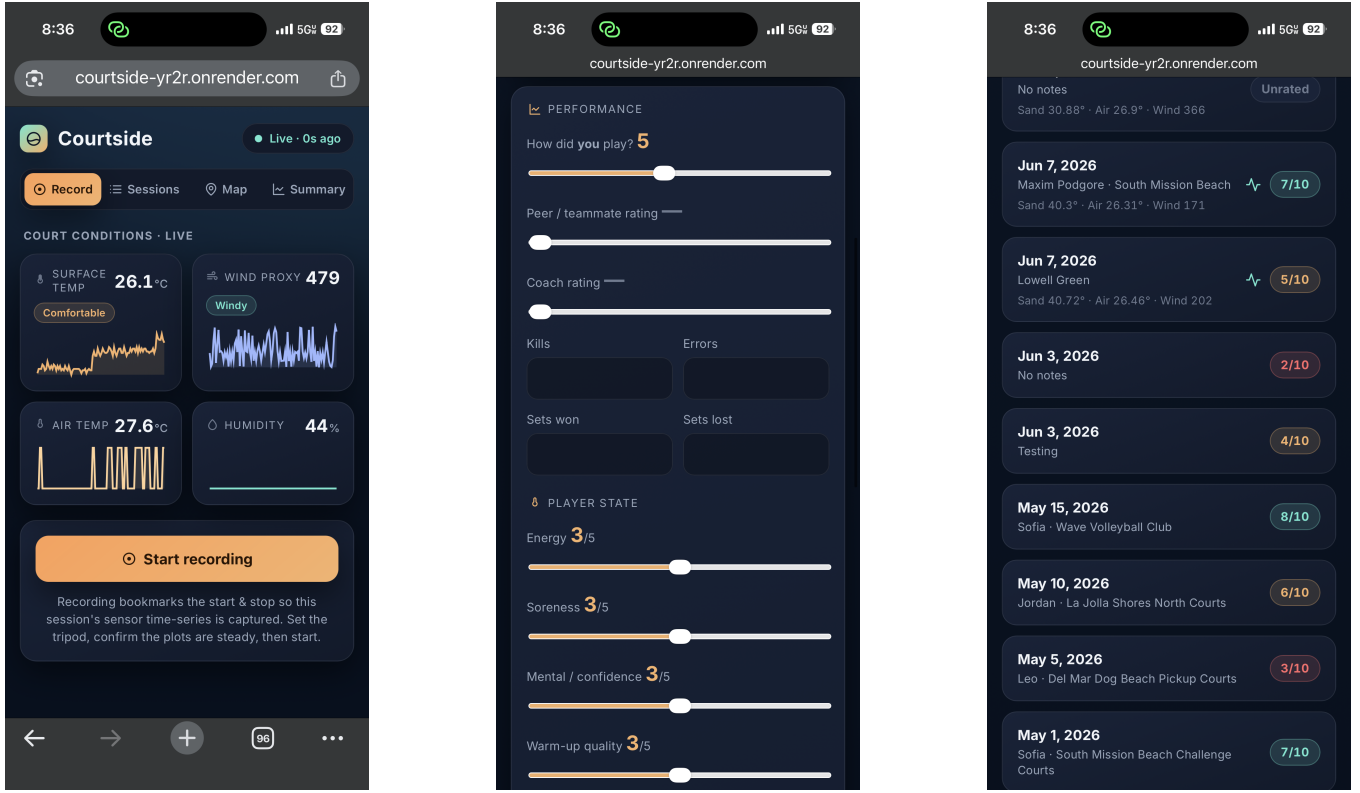


Figure 5: The deployed app on the athlete’s phone over 5G, with four tabs (Record, Sessions, Map, Summary). Left: the live setup view (“Live · 0s ago”) with balanced per-sensor cards and a “Comfortable/Windy” interpretation. Center: the post-session evaluation form, including the survey-driven player-state sliders (energy, soreness, mental, warm-up). Right: the session list—the June entries are real pilot sessions (sand 40.3–40.7 °C against ~26 °C air, workout-paired), interleaved with sample sessions used for demonstration.

Fabrication

The node was hand-built and hand-soldered (Fig. 8, left), with each subsystem brought up and smoke-tested individually before integration (Fig. 8, right shows the first working IR read).

6 Methodology

6.1 Data Handling and Feature Engineering

Sensor rows are stamped with server-receive time. Each logged session windows the node readings to its recording interval and aggregates them (mean/max sand and air temperature, mean humidity, mean and 95th-percentile wind-audio). Daily wearable recovery joins by date (Apple preferred, Oura fallback and cross-check). Apple Watch *workouts* overlapping the session window yield in-session intensity (average/max HR, active energy); workouts in the prior seven days yield a cross-training *load* feature. Regional weather is attached per session location, and partner/opponent links attach the relational context. The unit of analysis is the session—appropriate given small N —producing one tidy multi-modal feature row per session (~40 columns).

6.2 Analysis Pipeline

The analysis is exploratory and descriptive, in four small- N -aware stages: (1) Pearson and Spearman correlations of each feature against the subjective rating; (2) Lasso regression with leave-one-out cross-validation for sparse feature selection; (3) random-forest and gradient-boosting feature importances to capture interactions; (4) SHAP values [3] from the gradient-boosting model for per-factor attribution. We additionally compute leave-one-session-out top-3 stability. The pipeline (scikit-learn [4]) degrades gracefully: below a sample threshold it emits correlations plus an explicit caveat rather than fabricating a model. No p -values are reported as confirmatory, and no model is released for deployment.

6.3 Relationship Statistics

For the relational signals, each person’s with-set and without-set of ratings are compared with a two-sided Welch’s t -test (unequal variance), with the p -value computed from the regularized incomplete beta function in pure Python. Signals fire only with ≥ 3 sessions for the person, an effect of ≥ 0.8 rating points, and $p < 0.1$ —thresholds chosen to be informative on small samples while explicitly labeled preliminary.

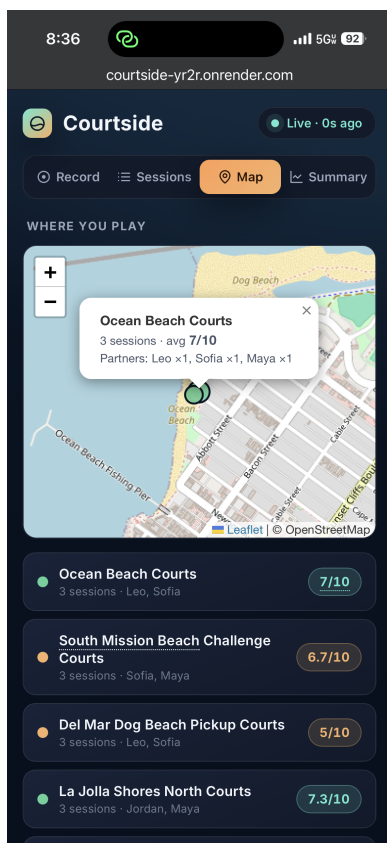


Figure 6: The Map view with a court tapped open: session count, average rating, and who the athlete has played with there, over markers colored by average rating. Populated mostly with sample data to demonstrate the feature; the pilot’s real courts appear here too.

6.4 Evaluation Criteria

Because the output is a per-session explanation rather than a prediction, standard ML metrics do not apply. We evaluate on engineering criteria: **data completeness** (fraction of sessions with all streams captured), **time alignment** (streams aligned to server-receive time and the recording window), **sensor responsiveness** (bench validation per channel), and **ranking stability / face validity** (leave-one-session-out stability and comparison of top factors to the athlete’s recollection)—the latter two meaningful only once N is sufficient.

7 Results

7.1 User-Need Validation (N=20)

Before further hardware investment we surveyed 20 players (UCSD men’s/women’s beach teams plus recreational, club, NCAA, and Open/AVP players) to validate need and prioritize features (Fig. 9). **Wind ranked first** (4.05/5), validating the decision to prioritize wind sensing; recovery/tiredness, mental state, partner identity, and partner communication completed the top five; court noise, location, and pace of play ranked lowest and were dropped from the feature plan. Open responses repeatedly raised *food timing*,

soreness, and *warm-up quality*, motivating the lightweight player-state inputs. Across the cohort, 85% expressed positive interest, with qualified interest centered on wearable integration and minimal in-session burden.

7.2 Sensor Validation (Bench)

Each channel was validated for responsiveness (Fig. 10): exhaling on the node produces the expected coupled DHT11 temperature/humidity spike with exponential return to ambient; blowing across the microphone produces clear transient amplitude spikes above a stable baseline; and the MLX90614 tracks a warm surface in its field of view (an immediate $\sim 10^\circ\text{C}$ rise aimed at skin). The Oura API pipeline was separately validated against a 21-day rolling window (Fig. 11).

7.3 Live Courtside Pilot

The full wireless path was exercised on the beach (Fig. 1, Fig. 5): the node ran from a USB battery, joined the phone hotspot, and streamed to the cloud while the athlete used the live dashboard to position the tripod. On instrumented beach days we recorded real sessions with sand-surface temperatures of $40.3\text{--}40.7^\circ\text{C}$ against $\sim 26^\circ\text{C}$ air—directly demonstrating the court-level microclimate that weather data cannot capture—tagged with real partners (e.g., a session at South Mission Beach) and real ratings. Apple Watch recovery for the same windows arrived automatically (HRV, resting HR, sleep), and Volleyball workouts were ingested with in-session heart rate (average 140–172 bpm, peak up to ~ 186 bpm) and active energy (160–459 kcal); Figure 12 shows this real wearable data populated live in the app. The build step assembled a ~ 40 -column multi-modal feature row per session, confirming end-to-end fusion. In total, the pilot produced **two fully-instrumented sessions** carrying every stream simultaneously—enough to prove the fusion end-to-end, but not enough for inferential ranking.

7.4 Pipeline Demonstration (Sample Data)

Because the number of *fully-instrumented* real sessions is currently too small for interpretable ranking—**only $N=2$ sessions to date carry every stream simultaneously**, and a correlation over a handful of points can exceed 0.8 by chance—we ran the unmodified pipeline on a clearly-labeled sample dataset of 16 sessions *built around those two real sessions*: it matches their observed value ranges (e.g. $\sim 40^\circ\text{C}$ sand and the measured heart-rate and sleep ranges) and adds a planted plausible structure (better sleep/HRV and cooler, calmer conditions $\rightarrow$ higher rating). The pipeline recovers that structure (Fig. 13): HRV correlates positively ($r \approx 0.67$) and sand-surface temperature negatively ($r \approx -0.36$) with the rating, and the leave-one-session-out stability column flags which factors are robust. This demonstrates *what the instrument produces at scale*, not a finding.

The instrument also produces targeted, personally-actionable relationship views (Fig. 14). Pairing each Apple Watch workout to its node session window by *timestamp overlap* yields *actual effort* (in-session heart rate) alongside *reported performance*; overnight sleep can likewise be plotted against performance. On sample data, sleep tracks self-rated performance more strongly than raw physical

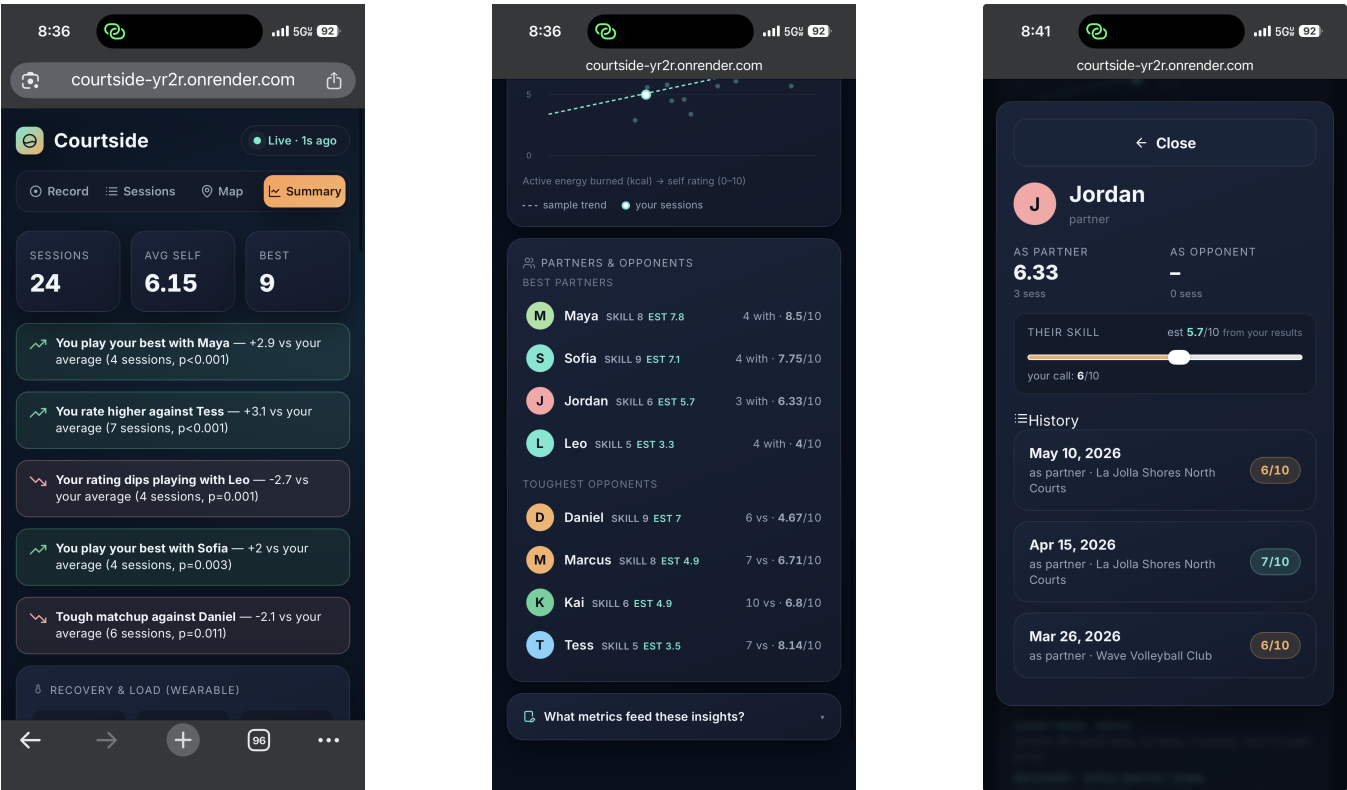


Figure 7: Relational analytics (sample data; the pilot has only $N=2$ fully-instrumented real sessions, so these views are demonstrations built around the pilot’s data). Left: relationship signals comparing the athlete’s ratings with/without each person (Welch’s t , small- N guarded)—partner boosts and the “tough matchup” opponent drags. Center: partners and opponents ranked, each with a self-rated and a results-estimated skill. Right: a person view (a partner here) with as-partner/as-opponent averages and full history.

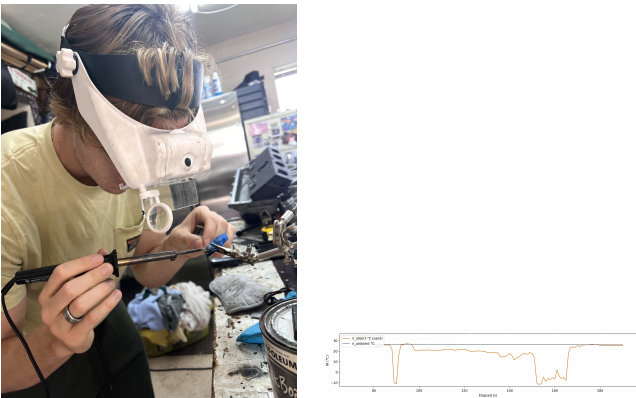


Figure 8: Fabrication and bring-up. Left: hand-soldering the node under a magnifier visor. Right: the first successful MLX90614 infrared read during bench bring-up.

effort—exactly the contrast the tool is designed to surface, and a reminder that more effort is not more performance.

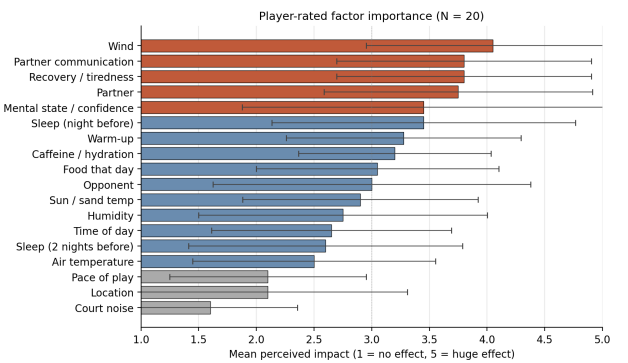


Figure 9: Player-rated factor importance ($N = 20$), mean $\pm$ 1 SD. Wind ranks first overall. Top five (orange) retained as primary engineered features; bottom three (gray) dropped.

8 Discussion

What worked. The hardest integration risks—untethered wireless streaming from a beach, automated wearable ingestion, and time-aligned multi-modal fusion across six streams—are all working

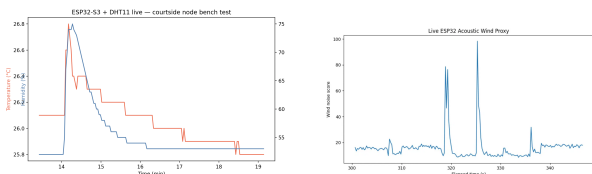


Figure 10: Bench validation. Left: an exhalation event produces a coupled temperature/humidity response. Right: the acoustic wind proxy responds to blown-air transients above a stable baseline.

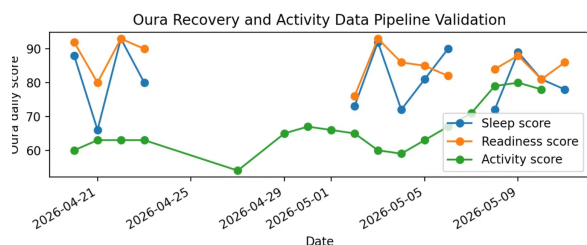


Figure 11: Oura recovery pipeline validated over a rolling multi-week window (HRV, resting HR, sleep), providing an independent cross-check on the Apple Watch recovery stream.

and deployed. The court-level microclimate signal ($\sim 40^\circ\text{C}$ sand vs. $\sim 26^\circ\text{C}$ air) is real and is precisely the value a single wearable or weather app misses.

Where the value lies. Given the sample size, we are explicit that the contribution is *not* statistical. The value of this project is (i) the technology—a genuinely working, deployed, multi-modal instrument; (ii) the user-needs study that grounds it; (iii) filling a real gap no existing tool addresses; and (iv) an extensible platform with clear potential. We make no inferential claims on the real data.

Honest limitations. *Small N is the dominant limitation:* factor ranking on the two fully-instrumented sessions to date is not interpretable, so all real output is framed descriptively and the pipeline reports its own instability. *Wind sensing is fundamentally hard with this hardware:* estimating true wind speed from a single small microphone behind a port is essentially not possible—a credible anemometric estimate would need a higher-quality sensor and likely several of them. The current proxy is therefore qualitative, validated against self-reported wind and regional weather rather than against a calibrated anemometer, and more field validation is needed. *Sensor placement is fixed* within the enclosure: the whole node can be rotated to face the wind correctly, but doing so also re-aims the IR sensor, a coupling a future revision should decouple. *Sensor fidelity:* the DHT11 is low-resolution ($\pm 2^\circ\text{C}$, $\pm 5\%$ RH, no pressure). *Ground truth:* the subjective 1–10 rating is the primary outcome; kills/errors are secondary and heavily confounded by partner and opponent in 2v2 play, which is exactly why we added peer/coach ratings and the relational analytics rather than over-trusting raw counts.

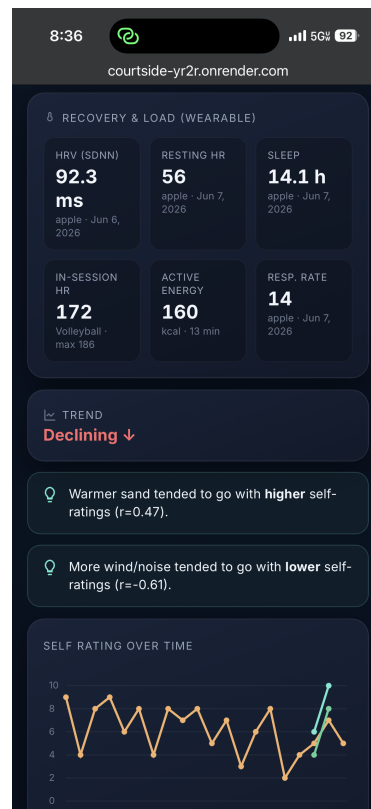


Figure 12: The Summary recovery panel with real pilot wearable data: HRV (SDNN), resting HR, sleep, the in-session heart rate from a timestamp-paired Volleyball workout (avg 172, max 186 bpm), and active energy—flowing automatically from the Apple Watch into the deployed app.

Scope cuts (and the roadmap). Two stretch goals were deliberately cut to keep the core robust, and both are planned as I continue this as a personal project. *Computer-vision performance metrics*—automated kills, digs, reach, vertical jump, and score tracking—are feasible with existing technology and mainly need integration. *Voice EMA / partner-communication capture* would lower in-session reporting burden. Neither was essential to proving the instrument.

Reviewer feedback addressed. Per TA and classmate feedback: the report is structured with clearly-marked sections and judicious, high-resolution figures; the survey is elevated as user-need validation; the kills/errors confound is reframed with peer/coach ground truth and relational context; an evaluation section reports what is measurable at current N plus a concrete plan; and the analysis is a committed, specified four-stage pipeline.

Generalization. The patterns—time-aligning heterogeneous low-cost streams, capturing microclimate that regional data misses, small- N feature ranking with honest stability reporting, and spatial/relational context—transfer to other outdoor, individually-variable, heat-exposed settings (e.g., heat-exposed workers).

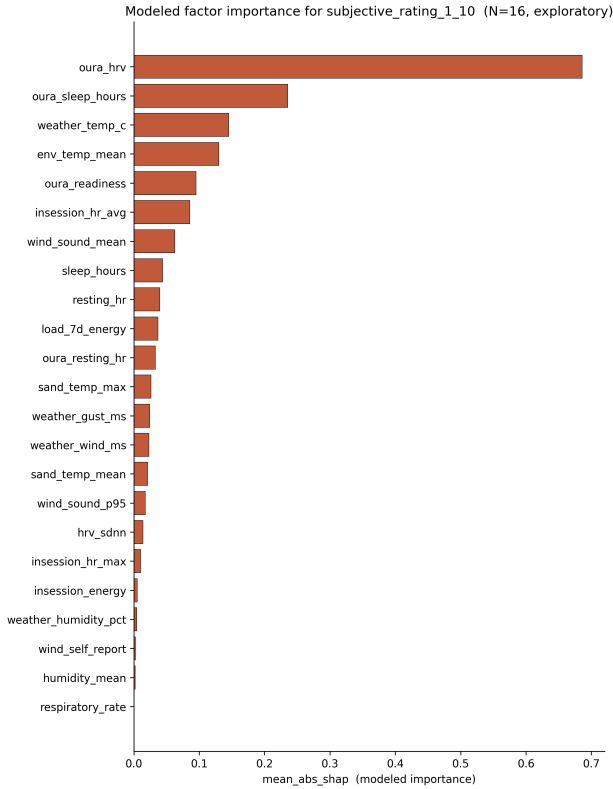


Figure 13: Pipeline factor-ranking output on a labeled sample 16-session dataset built around the pilot’s $N=2$ real sessions (not real results). Demonstrates the end-to-end analysis and its self-reported stability; real-data ranking awaits sufficient N .

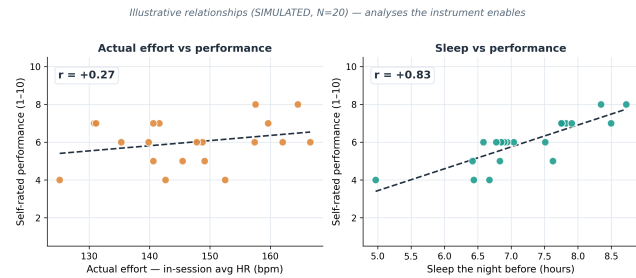


Figure 14: Relationship views the instrument produces from paired streams (sample data built around the pilot’s $N=2$ real sessions): actual effort (watch in-session HR, timestamp-matched to the node session) vs. self-rated performance (left), and sleep vs. performance (right). Shown to illustrate the analysis; not a real-data finding.

9 Conclusion

We delivered a working, deployed multi-modal instrument and analysis pipeline that fuses court-level environmental sensing, wearable recovery and in-session intensity, regional weather, spatial (court

and relational (partner/opponent) context, and subjective performance for an individual beach volleyball athlete—grounded in a 20-player needs survey and validated end-to-end on a live beach pilot. The main takeaway: the integration is real and the court microclimate it captures is genuinely invisible to existing tools; the honest gap is sample size, and because the instrument now collects all streams at near-zero marginal effort, the remaining work—reaching enough sessions and re-running the already-built ranking—is straightforward. We present sample-data output (built around the pilot’s $N=2$ real sessions) to show what the system produces at scale, and we make no statistical claims on the small real sample.

AI Use Disclosure

AI tools (Claude) were used as a pair-programming and drafting assistant for the firmware, backend, and web-app code, for editing this report’s prose, and to summarize open-ended survey responses. All system design, hardware integration, data-collection methodology, analysis choices, and the underlying data are the author’s own.

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